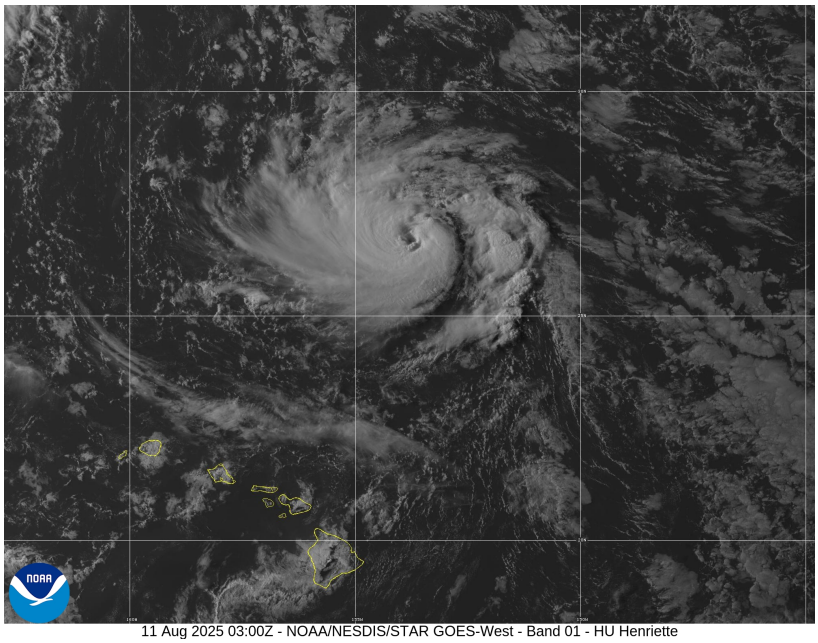
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE HENRIETTE

(EP082025)

4–13 August 2025

Alex Gibbs
National Hurricane Center
24 November 2025



GOES-WEST VISIBLE SATELLITE IMAGE OF HURRICANE HENRIETTE AT 0300 UTC 11 AUGUST 2025, NEAR THE TIME OF ITS PEAK INTENSITY. IMAGE COURTESY NOAA/NESDIS/RAMMB/CIRA.

Henriette was a hurricane that formed well southwest of the Baja California Peninsula and did not bring any direct impacts to land. The system briefly became a remnant low as it approached and crossed into the central Pacific basin, but later reorganized and intensified into a Category 1 hurricane as it passed well north of Hawaii.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Henriette.

Hurricane Henriette

4–13 August 2025

BEST TRACK

The "best track[2]" positions and intensities for Hurricane Henriette are listed in Table 1. The "best track" chart of Henriette's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no ship observations of tropical-storm-force winds associated with Henriette.

Origin

Henriette originated from a tropical wave that moved off the coast of Africa on 16 July and crossed Central America around 28–30 July. The system emerged over the eastern Pacific and continued westward for a few days, eventually evolving into a broad area of low pressure while producing disorganized convection. Showers and thunderstorms associated with this system gradually became better organized when the disturbance was located well southwest of the Baja California Peninsula on 3 August. It is estimated that the system became a tropical depression at 0600 UTC 4 August, when scatterometer data revealed a well-defined surface circulation and deep convection consolidated over the center. The cyclone strengthened into a 35-kt tropical storm about 6 h later.

Peak Intensity and Minimum Pressure

Henriette's estimated peak intensity of 75 kt around 1200 UTC 11 August is based on a blend of subjective Dvorak intensity estimates from HFO, JTWC, and SAB, along with objective estimates from UW-CIMSS. The estimated minimum central pressure of 986 mb is based primarily on the Knaff–Zehr–Courtney (KZC) pressure–wind relationship. It should be noted that

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Central Pacific Hurricane Center (PHFO), the Joint Typhoon Warning Center (JTWC), Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Henriette.

tropical cyclones reaching their peak intensity this far north of Hawaii, where cooler waters, drier air, and stronger vertical wind shear generally prevail, are rare.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Henriette.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The tropical wave that eventually became Henriette was introduced in the TWO with sufficient lead time, with low probabilities of formation (<30%) in the 2- and 7-day periods about 84 h and 144 h prior to genesis, respectively. These probabilities were gradually increased as the system began to consolidate and become more organized, with high formation probabilities (>60%) introduced about 30 h prior to genesis in the 2-day period and about 96 h prior in the 7-day period. Composites of the 7-day TWO genesis areas (Fig. 4) show that the spatial extent of the genesis region remained consistent through time, with the medium (40–60%) and high (>60%) probability contours centered very near the genesis location. As a result, the forecasts of Henriette's formation location were highly accurate, with the cyclone forming near the center of the depicted genesis areas, yielding a 100% hit rate. It should also be noted that while Henriette weakened into a post-tropical low and then re-intensified in the central Pacific, NHC continued to issue official forecasts, maintaining forecast continuity while the system did not meet the criteria of a tropical cyclone.

A verification of the NHC official track forecasts for Henriette is provided in Tables 3a and 5a covering the eastern and central North Pacific basins, respectively. In both basins, the official track forecast errors were lower than the mean official errors from the previous 5-year period at all forecast times. A homogeneous comparison of the official track errors with selected guidance models is provided in Tables 3b and 5b. Several models outperformed the official NHC forecast through 36 h in the eastern Pacific; however, from 48 h through 120 h, the official forecasts were generally more accurate than the majority of the guidance. In the central Pacific, the official forecasts were typically more accurate than the guidance models at most forecast times.

A verification of the NHC official intensity forecasts for Henriette is provided in Tables 4a and 6a covering the eastern and central North Pacific basins, respectively. In the eastern Pacific, the official intensity forecast errors were lower than the mean official errors from the previous 5-year period at all forecast times. In the central Pacific, the official errors were lower than the 5-year period from 60 h to 72 h. A homogeneous comparison of the official intensity errors with selected guidance models is provided in Tables 4b and 6b. Several models outperformed the official NHC forecasts during the 36–60 h forecast periods in the eastern Pacific. In the central

Pacific, several models outperformed the official forecasts through 60 h, with the FSSE and HFAI showing smaller errors than the official forecasts at all time periods.

There were no coastal watches or warnings issued for Henriette.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the Graphical TWO verification figure.

Table 1. Best track for Hurricane Henriette, 4–13 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
04 / 0600	13.4	117.5	1008	30	tropical depression
04 / 1200	14.1	118.8	1006	35	tropical storm
04 / 1800	14.6	120.0	1004	40	"
05 / 0000	15.1	121.2	1003	45	"
05 / 0600	15.7	122.4	1003	45	"
05 / 1200	16.3	123.7	1003	45	"
05 / 1800	16.8	124.9	1003	45	"
06 / 0000	17.4	126.1	1003	45	"
06 / 0600	17.7	127.4	1003	45	"
06 / 1200	17.9	128.9	1004	45	"
06 / 1800	18.0	130.3	1004	45	"
07 / 0000	18.1	131.7	1004	45	"
07 / 0600	18.3	133.1	1004	45	"
07 / 1200	18.5	134.6	1004	45	"
07 / 1800	18.7	136.1	1007	35	"
08 / 0000	18.9	137.7	1007	35	"
08 / 0600	19.1	139.3	1008	35	low
08 / 1200	19.5	140.8	1008	35	"
08 / 1800	19.8	142.3	1009	30	"
09 / 0000	20.0	143.9	1009	30	"
09 / 0600	20.4	145.1	1009	30	tropical depression

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
09 / 1200	21.1	146.3	1009	30	"
09 / 1800	21.8	147.5	1008	35	tropical storm
10 / 0000	22.6	148.6	1005	40	"
10 / 0600	23.4	149.8	1002	45	"
10 / 1200	24.2	150.9	998	55	"
10 / 1800	25.2	151.9	993	65	hurricane
11 / 0000	26.2	153.2	990	70	"
11 / 0600	27.2	154.4	990	70	"
11 / 1200	28.3	155.8	986	75	"
11 / 1800	29.3	157.2	986	75	"
12 / 0000	30.3	158.4	987	70	"
12 / 0600	31.3	159.6	990	65	"
12 / 1200	32.3	161.0	994	60	tropical storm
12 / 1800	33.6	162.3	999	55	"
13 / 0000	34.7	163.7	1001	50	"
13 / 0600	36.0	165.0	1006	40	low
13 / 1200	37.4	166.3	1011	35	"
13 / 1800	39.0	167.6	1014	30	"
14 / 0000	40.5	168.3	1014	30	"
14 / 0600	42.1	168.4	1016	25	"
14 / 1200	-	-	-	-	dissipated
11/1200	28.3	155.8	986	75	minimum pressure & maximum winds

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	84	144
Medium (40%-60%)	54	126
High (>60%)	30	96

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Henriette in the eastern North Pacific basin, 4–13 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**18.5**	**25.0**	**26.8**	**20.4**	**19.0**	**27.0**	**23.8**	**64.9**
OCD5	20.3	51.5	87.9	126.6	166.4	223.9	386.2	496.0
Forecasts	14	12	12	12	12	12	12	16
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Henriette in the eastern North Pacific basin, 4–13 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	18.5	26.1	27.1	20.7	18.8	26.9	23.2	69.8
OCD5	22.2	53.3	89.5	124.7	167.0	222.4	384.8	501.7
GFSI	**17.3**	**24.8**	**26.0**	21.4	26.5	45.5	58.9	71.4
HMNI	**17.9**	**24.6**	34.4	38.5	50.8	71.8	82.4	96.4
HWFI	23.2	37.1	33.9	29.7	41.7	51.6	91.7	152.1
EMXI	18.9	30.3	35.4	36.0	27.5	**25.5**	54.8	85.9
CMCI	**17.4**	**25.0**	**25.1**	24.6	30.0	38.7	69.7	108.7
NVGI	26.1	59.6	84.3	111.7	144.1	164.6	160.6	281.8
CTCI	**15.2**	**24.2**	**25.8**	**20.0**	19.2	33.1	78.4	147.5
AEMI	**18.1**	26.6	29.3	25.2	31.9	46.2	53.7	**65.3**
HCCA	**17.2**	**25.5**	**25.6**	24.2	27.0	31.3	28.5	**67.1**
FSSE	**16.3**	**24.6**	**26.6**	25.0	25.4	**23.4**	27.2	77.5
TVCX	**15.0**	**23.7**	**25.8**	22.6	21.6	28.9	32.8	**69.1**
GFEX	**16.6**	26.1	27.3	25.8	21.0	30.5	44.1	73.7
TVCE	**15.2**	**23.3**	**25.5**	22.3	21.6	30.3	32.4	70.6
TVDG	**15.3**	**22.9**	**26.0**	23.3	22.1	30.4	30.9	**65.7**
TABD	26.6	55.1	98.7	138.6	173.5	203.6	234.9	235.3
TABM	19.7	35.6	51.5	64.3	80.6	95.5	125.8	147.7
TABS	19.1	33.6	53.0	63.6	75.8	79.9	96.0	122.3
HFAI	**16.0**	26.9	34.5	38.9	39.3	40.3	34.9	81.9
HFBI	**16.2**	26.2	30.6	28.4	28.7	36.0	44.8	82.1
Forecasts	12	10	10	10	11	11	11	14

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Henriette in the eastern North Pacific basin, 4–13 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**2.9**	**3.8**	**5.4**	**6.7**	**6.2**	**6.7**	**7.1**	**13.1**
OCD5	3.4	4.5	5.8	6.8	11.4	14.9	23.8	21.9
Forecasts	14	12	12	12	12	12	12	16
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Henriette in the eastern North Pacific basin, 4–13 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.3	4.0	5.5	7.0	6.8	6.4	7.7	12.5
OCD5	3.8	4.7	5.7	**6.6**	11.3	14.3	25.6	22.4
GFSI	4.0	5.1	**5.2**	**5.3**	7.4	10.1	13.8	**11.6**
HMNI	4.5	5.1	**5.4**	**6.9**	9.9	9.9	**6.5**	14.4
HWFI	5.0	4.1	**3.0**	**3.5**	**6.5**	9.0	20.8	14.9
EMXI	3.8	4.4	**4.5**	**5.0**	7.4	9.5	12.8	**10.9**
CTCI	3.4	4.3	**3.5**	**4.7**	**5.9**	8.4	12.4	17.1
HCCA	3.9	6.1	10.0	7.9	**5.5**	**4.5**	9.0	**10.9**
FSSE	3.7	4.3	**5.4**	**6.8**	8.4	6.5	10.3	16.9
DSHP	4.2	**3.9**	**3.8**	**4.5**	8.5	15.1	28.5	28.3
LGEM	4.4	5.2	**5.0**	**5.8**	8.4	16.9	31.4	30.8
ICON	4.2	4.1	**3.2**	**3.7**	**4.4**	8.3	19.0	16.8
IVCN	3.8	4.0	**3.6**	**4.1**	**4.8**	**6.1**	14.0	**12.0**
IVDR	3.8	4.0	**4.1**	**4.8**	**5.5**	**5.8**	12.0	**11.6**
HFAI	3.8	**3.7**	**4.9**	8.2	10.8	8.2	9.8	12.5
HFBI	4.7	4.8	7.4	10.3	12.5	9.0	13.8	17.6
Forecasts	12	10	10	10	11	11	11	14

Table 5a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Henriette in the central North Pacific basin, 4–13 August 2025. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**12.4**	**16.4**	**16.8**	**15.7**	**28.8**	**31.1**		
OCD5	31.1	87.3	178.0	284.3	388.7	485.8		
Forecasts	14	12	10	8	6	4		
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4		
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0		

Table 5b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Henriette in the central North Pacific basin, 4–13 August 2025. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	12.4	16.4	16.8	15.7	28.8	31.1		
OCD5	31.1	87.3	178.0	284.3	388.7	485.8		
GFSI	13.1	17.0	17.8	19.5	30.6	33.2		
HMNI	15.8	32.0	38.8	51.5	83.0	96.2		
HWFI	15.0	28.6	37.9	46.5	73.6	136.5		
EMXI	13.4	26.0	32.6	36.4	57.9	47.3		
CMCI	15.0	22.4	26.9	41.6	68.6	78.7		
NVGI	18.3	36.6	57.1	84.8	103.8	123.7		
CTCI	16.2	29.2	43.8	73.3	114.2	162.6		
AEMI	14.2	21.8	27.1	32.6	47.8	73.9		
HCCA	13.2	23.5	27.2	34.1	38.4	40.6		
FSSE	13.9	21.3	29.4	40.5	47.8	53.7		
TVCX	**12.2**	19.3	21.6	23.1	39.0	47.7		
GFEX	**11.1**	16.9	21.6	20.8	39.9	31.3		
TVCE	12.5	20.1	21.3	22.6	40.3	50.9		
TVDG	12.5	18.5	20.6	19.9	34.4	38.4		
TABD	23.3	47.7	71.4	107.9	152.7	196.6		
TABM	22.2	44.3	69.5	100.7	144.2	189.3		
TABS	18.6	37.5	55.3	75.2	97.5	123.3		
HFAI	13.4	21.1	25.4	31.3	30.2	36.7		
HFBI	14.6	22.0	22.1	25.5	35.2	57.7		
GDMI	**8.9**	17.5	20.9	19.3	33.7	49.2		
Forecasts	14	12	10	8	6	4		

Table 6a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Henriette in the central North Pacific basin, 4–13 August 2025. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	9.3	16.7	20.0	16.2	**10.0**	**5.0**		
OCD5	12.7	22.8	38.4	39.2	47.0	42.0		
Forecasts	14	12	10	8	6	4		
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5		
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3		

Table 6b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Henriette in the central North Pacific basin, 4–13 August 2025. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	9.3	16.7	20.0	16.2	10.0	5.0		
OCD5	12.7	22.8	38.4	39.2	47.0	42.0		
GFSI	**8.0**	**13.1**	**17.0**	17.2	13.8	13.5		
HMNI	11.2	18.8	20.8	20.5	16.0	14.2		
HWFI	**7.8**	**13.8**	**17.8**	16.2	**9.7**	7.2		
EMXI	**8.5**	**14.9**	**19.3**	20.9	18.2	15.5		
CTCI	**7.3**	**11.1**	**14.6**	**15.0**	**9.8**	6.8		
HCCA	**6.4**	**11.2**	**13.8**	**16.1**	10.8	**4.0**		
FSSE	**7.2**	**12.2**	**16.2**	**13.5**	**6.8**	**2.2**		
DSHP	9.3	**16.3**	21.5	23.1	20.5	17.5		
LGEM	9.6	**16.6**	22.5	27.1	30.7	29.8		
ICON	**9.1**	**16.0**	20.5	21.2	18.0	13.2		
IVCN	**7.7**	**12.9**	**16.6**	**15.9**	12.2	6.0		
IVDR	**7.4**	**12.5**	**15.8**	**14.5**	10.7	**4.8**		
HFAI	**7.0**	**10.2**	**11.8**	**7.8**	**5.0**	**4.0**		
HFBI	**6.1**	**7.4**	**7.8**	**8.1**	**7.2**	7.0		
GDMI	**6.4**	**10.8**	**13.1**	**14.6**	15.0	12.8		
Forecasts	14	12	10	8	6	4		

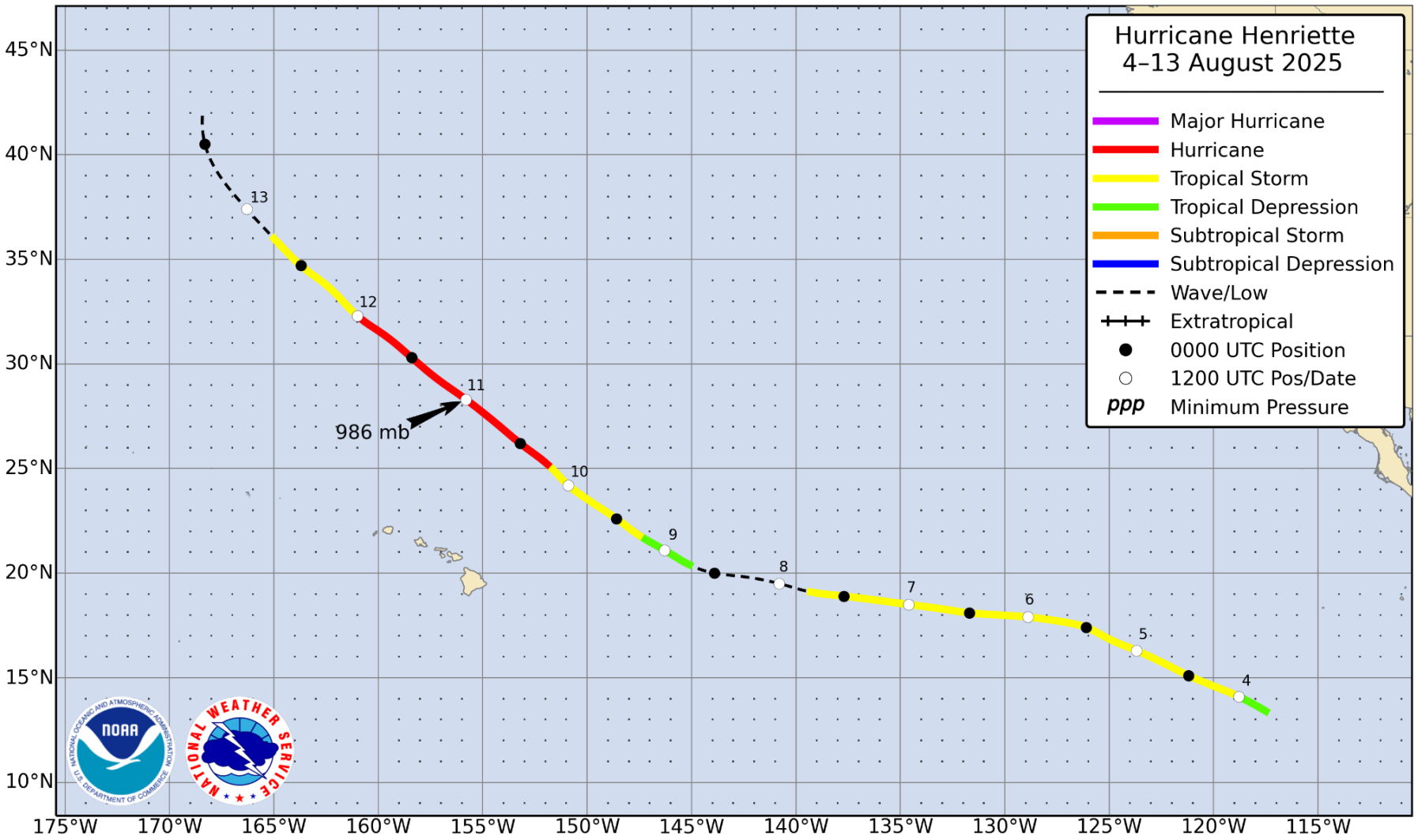


Figure 1. Best track positions for Hurricane Henriette, 4–13 August 2025.

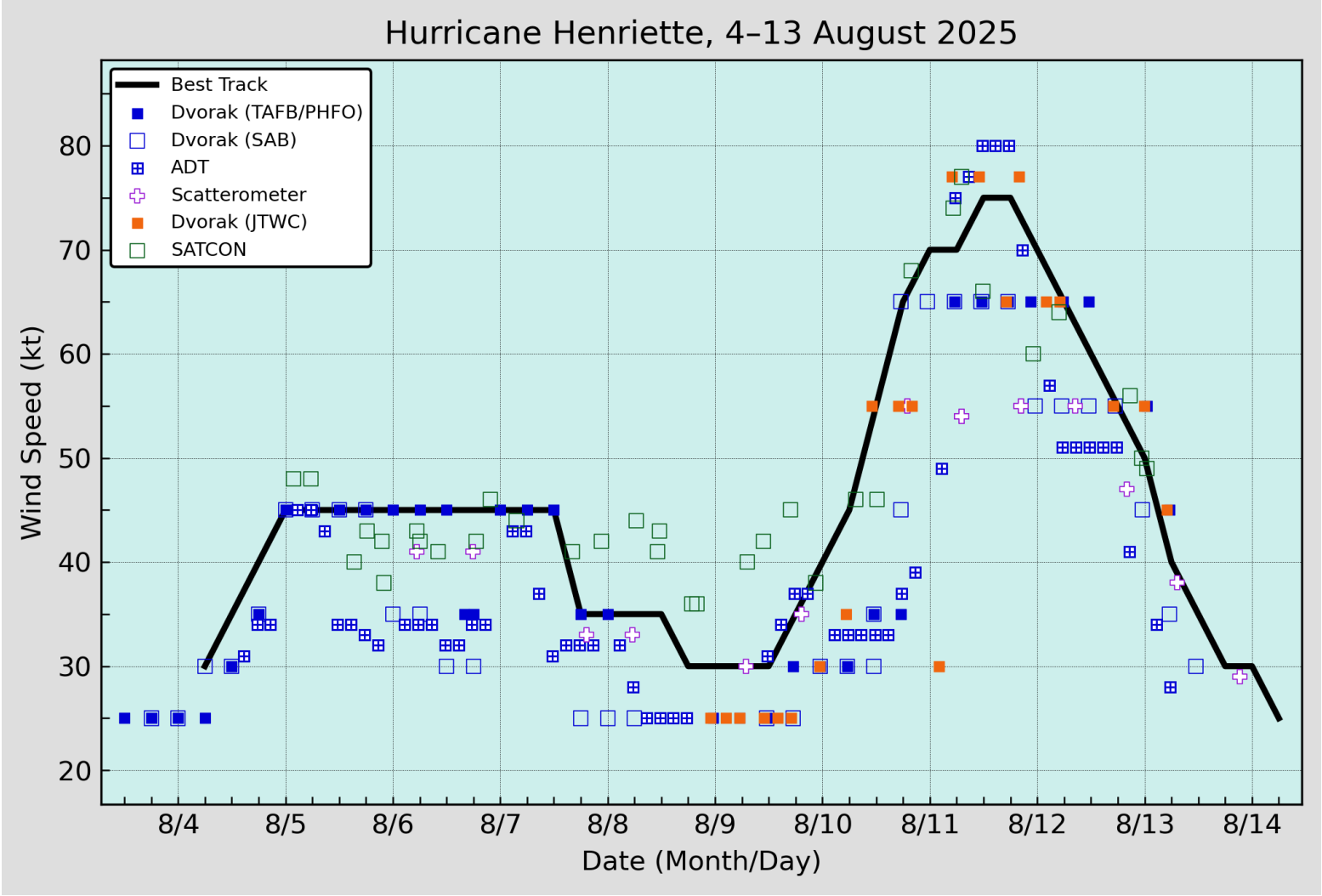


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Henriette, 4–13 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

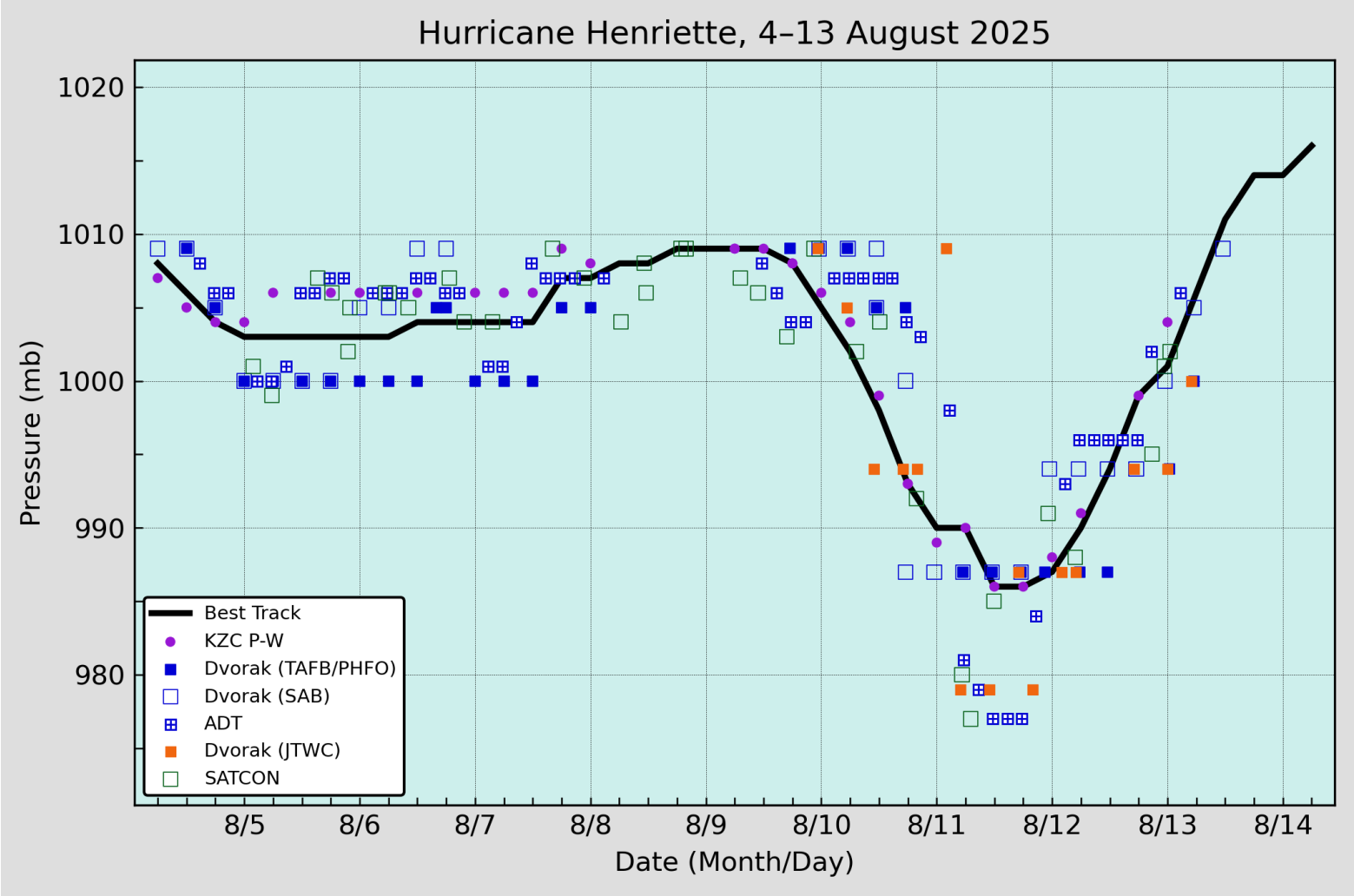


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Henriette, 4–13 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff–Zehr–Courtney (KZC) pressure–wind relationship. Dashed vertical lines correspond to 0000 UTC.

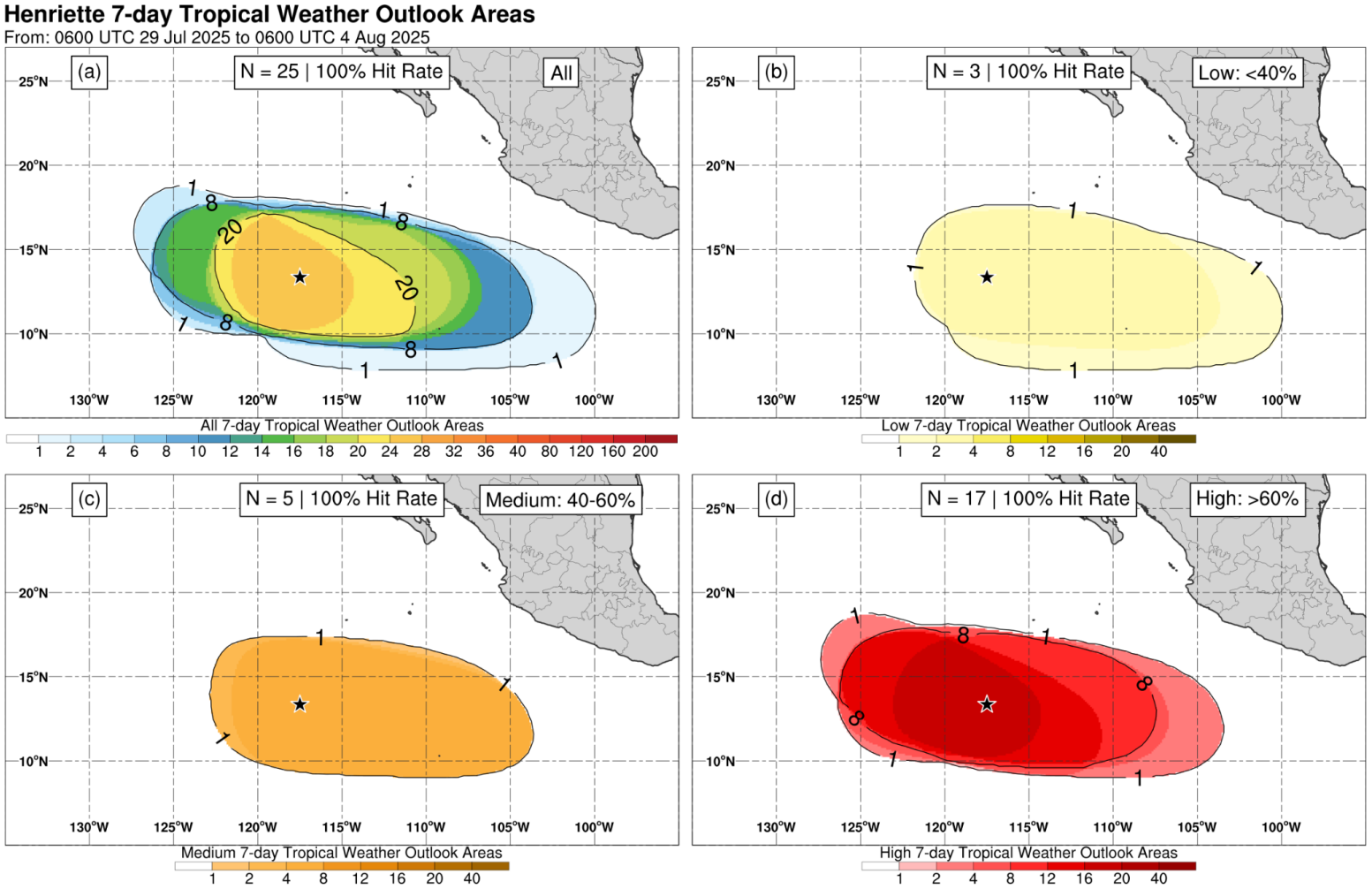


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Henriette for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.